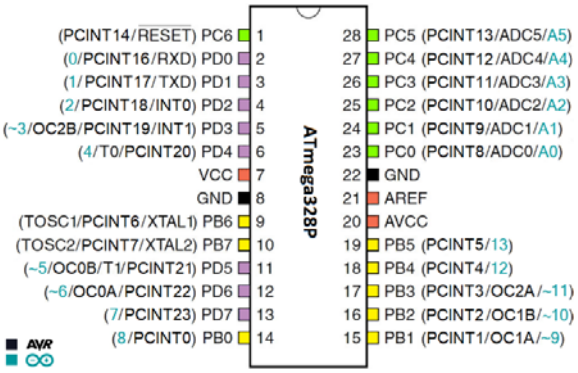
ATmega328P SIMPLIFIED DATA SHEET



I/O

Data Direction Register of port B (DDRB)				
Bit	Label	Description	R/W	DV
0	DDB0	DDRB bit 0	R/W	0
1	DDB1	DDRB bit 1	R/W	0
2	DDB2	DDRB bit 2	R/W	0
3	DDB3	DDRB bit 3	R/W	0
4	DDB4	DDRB bit 4	R/W	0
5	DDB5	DDRB bit 5	R/W	0
6	DDB6	DDRB bit 6	R/W	0
7	DDB7	DDRB bit 7	R/W	0

Data Direction Register of port C (DDRC)				
Bit	Label	Description	R/W	DV
0	DDC0	DDRC bit 0	R/W	0
1	DDC1	DDRC bit 1	R/W	0
2	DDC2	DDRC bit 2	R/W	0
3	DDC3	DDRC bit 3	R/W	0
4	DDC4	DDRC bit 4	R/W	0
5	DDC5	DDRC bit 5	R/W	0
6	DDC6	DDRC bit 6	R/W	0

Data Direction Register of port D (DDRD)				
Bit	Label	Description	R/W	DV
0	DDD0	DDRD bit 0	R/W	0
1	DDD1	DDRD bit 1	R/W	0
2	DDD2	DDRD bit 2	R/W	0
3	DDD3	DDRD bit 3	R/W	0
4	DDD4	DDRD bit 4	R/W	0
5	DDD5	DDRD bit 5	R/W	0
6	DDD6	DDRD bit 6	R/W	0
7	DDD7	DDRD bit 7	R/W	0

Port Output RegisTer B (PORTB)				
Bit	Label	Description	R/W	DV
0	PORTB0	PORTB bit 0	R/W	0
1	PORTB1	PORTB bit 1	R/W	0
2	PORTB2	PORTB bit 2	R/W	0
3	PORTB3	PORTB bit 3	R/W	0
4	PORTB4	PORTB bit 4	R/W	0
5	PORTB5	PORTB bit 5	R/W	0
6	PORTB6	PORTB bit 6	R/W	0
7	PORTB7	PORTB bit 7	R/W	0

Port Output RegisTer C (PORTC)				
Bit	Label	Description	R/W	DV
0	PORTC0	PORTC bit 0	R/W	0
1	PORTC1	PORTC bit 1	R/W	0
2	PORTC2	PORTC bit 2	R/W	0
3	PORTC3	PORTC bit 3	R/W	0
4	PORTC4	PORTC bit 4	R/W	0
5	PORTC5	PORTC bit 5	R/W	0
6	PORTC6	PORTC bit 6	R/W	0

Port Output RegisTer D (PORTD)				
Bit	Label	Description	R/W	DV
0	PORTD0	PORTD bit 0	R/W	0
1	PORTD1	PORTD bit 1	R/W	0
2	PORTD2	PORTD bit 2	R/W	0
3	PORTD3	PORTD bit 3	R/W	0
4	PORTD4	PORTD bit 4	R/W	0
5	PORTD5	PORTD bit 5	R/W	0
6	PORTD6	PORTD bit 6	R/W	0
7	PORTD7	PORTD bit 7	R/W	0

Port INput B (PINB) register				
Bit	Label	Description	R/W	DV
0	PINB0	PINB bit 0	R/W	0
1	PINB1	PINB bit 1	R/W	0
2	PINB2	PINB bit 2	R/W	0
3	PINB3	PINB bit 3	R/W	0
4	PINB4	PINB bit 4	R/W	0
5	PINB5	PINB bit 5	R/W	0
6	PINB6	PINB bit 6	R/W	0
7	PINB7	PINB bit 7	R/W	0

Port INput C (PINC) register				
Bit	Label	Description	R/W	DV
0	PINC0	PINC bit 0	R/W	0
1	PINC1	PINC bit 1	R/W	0
2	PINC2	PINC bit 2	R/W	0
3	PINC3	PINC bit 3	R/W	0
4	PINC4	PINC bit 4	R/W	0
5	PINC5	PINC bit 5	R/W	0
6	PINC6	PINC bit 6	R/W	0

Port INput D (PIND) register				
Bit	Label	Description	R/W	DV
0	PIND0	PIND bit 0	R/W	0
1	PIND1	PIND bit 1	R/W	0
2	PIND2	PIND bit 2	R/W	0
3	PIND3	PIND bit 3	R/W	0
4	PIND4	PIND bit 4	R/W	0
5	PIND5	PIND bit 5	R/W	0
6	PIND6	PIND bit 6	R/W	0
7	PIND7	PIND bit 7	R/W	0

INTERRUPTS

Commonly used Interrupt Vectors	
Interrupt	Vector
External interrupt on INT0 pin	INT0_vect
External interrupt on INT1 pin	INT1_vect
Pin Change INTerrupt on Port B	PCINT0_vect
Pin Change INTerrupt on Port C	PCINT1_vect
Pin Change INTerrupt on Port D	PCINT2_vect
ADc Complete	ADC_vect
TIMER/counter 2 OVerFlow	TIMER2_OVF_vect
TIMER/counter 1 OVerFlow	TIMER1_OVF_vect
TIMER/counter 0 OVerFlow	TIMER0_OVF_vect
USART Receive complete	USART_RX_vect
USART Transmit complete	USART_TX_vect
USART Data Register Empty	USART_UDRE_vect

External Interrupt Control Register A (EICRA)				
Bit	Label	Description	R/W	DV
0	ISC00	Int. Sense Control 00	R/W	0
1	ISC01	Int. Sense Control 01	R/W	0
2	ISC10	Int. Sense Control 10	R/W	0
3	ISC11	Int. Sense Control 11	R/W	0

ISC01	ISC00	Interrupt on
0	0	Low level of INT0 pin
0	1	Logic change of INT0 pin
1	0	Falling edge of INT0 pin
1	1	Rising edge of INT0 pin

ISC11	ISC10	Interrupt on
0	0	Low level of INT1 pin
0	1	Logic change of INT1 pin
1	0	Falling edge of INT1 pin
1	1	Rising edge of INT1 pin

External Interrupt MaSK (EIMSK) register				
Bit	Label	Description	R/W	DV
0	INT0	INTerrupt 0 enable	R/W	0
1	INT1	INTerrupt 1 enable	R/W	0

Pin Change Interrupt Control Register (PCICR)				
Bit	Label	Description	R/W	DV
0	PCIE0	Pin Change Interrupts Enable 0 on Port B	R/W	0
1	PCIE1	Pin Change Interrupts Enable 1 on Port C	R/W	0
2	PCIE2	Pin Change Interrupts Enable 2 on Port D	R/W	0

Pin Change MaSK 0 (PCMSK0) register				
Bit	Label	Description	R/W	DV
0	PCINT0	Pin Change INTerrupt 0 enable	R/W	0
1	PCINT1	Pin Change INTerrupt 1 enable	R/W	0
2	PCINT2	Pin Change INTerrupt 2 enable	R/W	0
3	PCINT3	Pin Change INTerrupt 3 enable	R/W	0
4	PCINT4	Pin Change INTerrupt 4 enable	R/W	0
5	PCINT5	Pin Change INTerrupt 5 enable	R/W	0
6	PCINT6	Pin Change INTerrupt 6 enable	R/W	0
7	PCINT7	Pin Change INTerrupt 7 enable	R/W	0

Pin Change MaSK 1 (PCMSK1) register				
Bit	Label	Description	R/W	DV
0	PCINT8	Pin Change INTerrupt 8 enable	R/W	0
1	PCINT9	Pin Change INTerrupt 9 enable	R/W	0
2	PCINT10	Pin Change INTerrupt 10 enable	R/W	0
3	PCINT11	Pin Change INTerrupt 11 enable	R/W	0
4	PCINT12	Pin Change INTerrupt 12 enable	R/W	0
5	PCINT13	Pin Change INTerrupt 13 enable	R/W	0
6	PCINT14	Pin Change INTerrupt 14 enable	R/W	0

Pin Change MaSK 2 (PCMSK2) register				
Bit	Label	Description	R/W	DV
0	PCINT16	Pin Change INTerrupt 16 enable	R/W	0
1	PCINT17	Pin Change INTerrupt 17 enable	R/W	0
2	PCINT18	Pin Change INTerrupt 18 enable	R/W	0

Pin Change MaSK 2 (PCMSK2) register				
Bit	Label	Description	R/W	DV

3	PCINT19	Pin Change INTerrupt 19 enable	R/W	0
4	PCINT20	Pin Change INTerrupt 20 enable	R/W	0
5	PCINT21	Pin Change INTerrupt 21 enable	R/W	0
6	PCINT22	Pin Change INTerrupt 22 enable	R/W	0
7	PCINT23	Pin Change INTerrupt 23 enable	R/W	0

ADC

ADC Low (ADCL) data register				
Bit	Label	Description	R/W	DV
0	ADCL0	ADC result bit 0	R	0
1	ADCL1	ADC result bit 1	R	0
2	ADCL2	ADC result bit 2	R	0
3	ADCL3	ADC result bit 3	R	0
4	ADCL4	ADC result bit 4	R	0
5	ADCL5	ADC result bit 5	R	0
6	ADCL6	ADC result bit 6	R	0
7	ADCL7	ADC result bit 7	R	0

ADC High (ADCH) data register				
Bit	Label	Description	R/W	DV
0	ADCH0	ADC result bit 8	R	0
1	ADCH1	ADC result bit 9	R	0

ADc MUltipleXer selection (ADMUX) register				
Bit	Label	Description	R/W	DV
0	MUX0	MUltipleXer selection 0	R/W	0
1	MUX1	MUltipleXer selection 1	R/W	0
2	MUX2	MUltipleXer selection 2	R/W	0
3	MUX3	MUltipleXer selection 3	R/W	0
5	ADLAR	ADc Left Adjust Result	R/W	0
6	REFS0	REFerence Selection 0	R/W	0
7	REFS1	REFerence Selection 1	R/W	0

MUX3	MUX2	MUX1	MUX0	Channel
0	0	0	0	ADC0
0	0	0	1	ADC1
0	0	1	0	ADC2
0	0	1	1	ADC3
0	1	0	0	ADC4
0	1	0	1	ADC5
1	0	0	0	Int. temp. sensor

ADLAR	ADC result in ADCH and ADCL
0	Right justified
1	Left justified

REFS1	REFS0	Vref
0	0	Set ARef pin externally
0	1	Set as AVcc
1	0	Reserved
1	1	Set as 1.1V internally

ADc Control and Status Register A (ADCSRA)				
Bit	Label	Description	R/W	DV
0	ADPS0	ADc Prescaler Select 0	R/W	0
1	ADPS1	ADc Prescaler Select 1	R/W	0
2	ADPS2	ADc Prescaler Select 2	R/W	0
3	ADIE	ADc Interrupt Enable	R/W	0
4	ADIF	ADc Interrupt Flag	R/W	0
5	ADATE	ADc Auto Trigger Enable	R/W	0
6	ADSC	ADc Start Conversion	R/W	0
7	ADEN	ADc Enable	R/W	0

ADPS0	ADPS0	ADPS0	F_ADC
0	0	0	Reserved
0	0	1	F_CPU/2
0	1	0	F_CPU/4
0	1	1	F_CPU/8
1	0	0	F_CPU/16
1	0	1	F_CPU/32
1	1	0	F_CPU/64
1	1	1	F_CPU/128

ADC Word (ADCW) register	
Description	10-bit ADC result

TIMER/COUNTER/PWM

Timer/Counter Control Register n A (TCCRnA)				
Bit	Label	Description	R/W	DV
0	WGMn0	Waveform Generation Mode n 0	R/W	0
1	WGMn1	Waveform Generation Mode n 1	R/W	0
4	COMnB0	Compare Output Mode n B0	R/W	0
5	COMnB1	Compare Output Mode n B1	R/W	0
6	COMnA0	Compare Output Mode n A0	R/W	0
7	COMnA1	Compare Output Mode n A1	R/W	0

COMnX1	COMnX0	Mode		OCnX Pin
0	0	Normal		Disconnected
1	0	Non-inverted	Fast FPWM	Clears on compare match & sets at bottom
			Phase Correct PWM	Clears on compare match when increment & sets on compare match when decrement
1	1	Inverted	Fast PWM	Sets on compare match & clears at bottom
			Phase Correct PWM	Sets on compare match when increment & clears on compare match when decrement

Timer/Counter Control Register n B (TCCRnB)				
Bit	Label	Description	R/W	DV
0	CSn0	Clock Select n 0	R/W	0
1	CSn1	Clock Select n 1	R/W	0
2	CSn2	Clock Select n 2	R/W	0
3	WGMn2	Waveform Generation Mode n 2	R/W	0
6	FOCnB	Force Output Compare n B	R/W	0
7	FOCnA	Force Output Compare n A	R/W	0

CSn2	CSn1	CSn0	Clock Source	
0	0	0	Disconnected	
0	0	1	F_CPU	
0	1	0	F_CPU/8	
0	1	1	F_CPU/64	
1	0	0	F_CPU/256	
1	0	1	F_CPU/1024	
1	1	0	External source on Tn pin	Counts on falling edge
1	1	1		Counts on rising edge

Output Compare Register n X (OCRnX)	
Description	Output compare value (X = A or B)

WGMn2	WGMn1	WGMn0	Mode
0	0	0	Normal
0	0	1	Phase Correct PWM
0	1	0	CTC
0	1	1	Fast PWM

Timer Interrupt Flag Register n (TIFRn)				
Bit	Label	Description	R/W	DV
0	TOVn	Timer OVerflow n flag	R/W	0
1	OCFnA	Output Compare match Flag n A	R/W	0
2	OCFnB	Output Compare match Flag n B	R/W	0

Timer Interrupt MaSK Register n (TIMSKn)				
Bit	Label	Description	R/W	DV
0	TOIEn	Timer Overflow Interrupt Enable n	R/W	0
1	OCIEnA	Output Compare match Interrupt Enable n A	R/W	0
2	OCIEnB	Output Compare match Interrupt Enable n B	R/W	0

Timer/CouNTer n (TCNTn) register	
Timer 0 / Timer 2	8-bit
Timer 1	16-bit

SERIAL COMMUNICATION

USART Data Register n (UDRn)	
Description	Transmit/receive data buffers

USART Baud Rate Register n (UBRRn)	
Description	[F_CPU / (16 x baud rate)] - 1

USART Control and Status Register n A (UCSRnA)				
Bit	Label	Description	R/W	DV
5	UDREn	USART Data Register Empty n flag	R	1
6	TXCn	Transmit Complete n flag	R/W	0
7	RXCn	Receive Complete n flag	R	0

USART Control and Status Register n B (UCSRnB)				
Bit	Label	Description	R/W	DV
2	UCSZn2	USART Character SiZe n 2	R/W	0
3	TXENn	Transmitter Enable n	R/W	0
4	RXENn	Receiver Enable n	R/W	0
5	UDRIEn	USART Data Register empty Interrupt Enable n	R/W	0
6	TXCIEn	Transmit Complete Interrupt Enable n	R/W	0
7	RXCIEn	Receive Complete Interrupt Enable n	R/W	0

USART Control and Status Register n C (UCSRnC)				
Bit	Label	Description	R/W	DV
1	UCSZn0	USART Character SiZe n 0	R/W	1
2	UCSZn1	USART Character SiZe n 1	R/W	1

UCSZn2	UCSZn1	UCSZn0	Character Size
0	0	0	5-bit
0	0	1	6-bit
0	1	0	7-bit
0	1	1	8-bit

Note
Unlike other bits, set flag bits to clear them

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